Building Materials

7 DETERMINE THE NORMAL CONSISTENCY FOR THE GIVEN SAMPLE OF CEMENT

Aim

To find out the normal consistency value for the given sample of cement.

Apparatus and Materials Required

TIZXLF

1. Vicat Apparatus
2. 10mm Dia Plunger
3. Mould
4. Non Porous Plate
5. Weighing Balance
6. Measuring Jar and Water
7. A Container
8. Trowel
9. Clock
10. Cement Sample (200 grams)

Procedure

1. Take 200 grams of cement.
2. Fix the plunger with the vicat apparatus and adjust the pointer to pointout zero on the vicat's scale when the plunger is placed above the non porous plate.
3. Add 20% of water [(20/100) × 200 = 40 ml] to the cement and prepare the paste.
4. Fill the prepared paste in the mould while the mould is placed above the non porous plate.
5. Bring the plunger just above the paste and allow it to penetrate the paste freely.
6. Note down weather the pointer points towards 5mm to 7mm on the vicat's scale.
7. If not, repeat the above steps by increasing the percentage of water upto which the pointer points 5mm to 7mm on the scale.
8. That is the percentage of normal consistency for that cement sample.

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